

Erdős Problem #445

EP-445

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Is it true that, for any $c > 1/2$, if p is a sufficiently large prime then, for any $n \geq 0$, there exist $a, b \in (n, n + p^c)$ such that $ab \equiv 1 \pmod{p}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>